

The Water-Energy Nexus

by **Ronnie Cohen**

It's not a secret, but most people don't think about it. Once you know, however, it seems obvious. Water uses a tremendous amount of energy. It is not just a matter of the gas and electricity required to heat, cool, or pump water in our homes and businesses. It takes large amounts of energy before that to extract, convey, treat, and deliver water. Then it takes more energy to collect, treat, and dispose of wastewater. In fact, the California Energy Commission estimates that almost 20% of California's electricity use, and over 30% of its natural gas use, is associated with the use of water.

Given these high energy requirements, and the way in which such energy use contributes to global climate change, perhaps we need to rethink our approach to water supply. Water and energy should be as closely linked in people's minds as peanut butter and jelly. Yet most people, and most policymakers, do not make this connection.

POWERFUL DEMAND

The more than 60,000 water systems and 15,000 wastewater systems in the United States are among the country's largest energy consumers. They use about 75 billion kilowatt-hours per year—3% of annual U.S. electricity consumption. This demand is equivalent to the entire residential demand for the state of California. And it doesn't even include the energy that is used to treat, circulate, heat, and cool water at the consumer level. Even if all of this power came from relatively clean, modern, natural-gas-fired power plants—which it doesn't—producing it would release about 30 million tons



Energy is needed at every stage in the water use cycle, including transporting it for storage. Water gets pumped uphill to this reservoir in Hollywood, California.

of carbon dioxide—more than the amount produced by 4 million cars.

Averages, of course, rarely tell the whole story. The energy required to use water varies greatly by location. In some states, such as California, where water is moved across long distances and carried over mountains, the amount of energy embedded in water can be significantly higher than average. The California State Water Project is the largest single user of energy in California. In the process of delivering water from the San Francisco Bay-Delta to Southern California, the project uses 2%–3% of all of the electricity consumed in the state. It takes an average of 3,000 kilowatt-hours per acre-foot to deliver water from the Bay-Delta to Southern California. (One acre-foot equals 325,000 gallons—roughly enough to meet the needs of two families of four for a year.) The state water project burns energy pumping water 2,000 feet over the Tehachapi Mountains—the highest lift of any water system in the world.

The Thirst for Energy. Why is water so energy intensive? The energy requirements of a given water supply reflect energy used at five stages of the water use cycle. Energy is needed to extract and convey water; to treat it; to distribute it locally; to use it; and to treat and dispose of the wastewater.

Extracting and conveying water. Most water used in the United States is diverted from surface sources, such as rivers and streams, or pumped from groundwater aquifers. This water may be used near its source, or it may be transported for use elsewhere. Conveying water often means pumping it over hills and mountains or into storage facilities—a process that can use many kilowatt-hours per acre-foot. These pumping systems usually recover some of the energy used to pump the water up and over the mountains with strategically placed hydroelectric generators that generate electricity as the water falls down the other side of those mountains.

Smaller amounts of fresh water are extracted from salt water, brackish

water, or wastewater, using desalination or water-recycling technologies. Desalination requires energy to remove salts from water, using reverse osmosis or other processes. Water recycling requires energy to remove pollutants from wastewater.

Treating water. Water treatment facilities use energy to pump and process water. The amount of energy needed for treatment depends on the quality of the source water. High-quality groundwater may need little treatment; surface water taken from rivers where wastewater is discharged upstream may need more. The energy needed to treat water is expected to increase over the next decade as treatment capacity expands, new water quality standards are put in place, and new treatments are developed to improve the taste and color of drinking water.

Distributing water. Energy is usually needed to pump and pressurize water, but gravity pressurization and distribution are also possible when reservoirs are sufficiently higher than the residences and businesses that will be using the water.

Using water. End users consume energy by further treating water with softeners or filters. They consume energy by circulating and pressurizing water with building circulation pumps and irrigation systems. Finally, end users consume energy by heating and cooling water.

Collecting and treating wastewater. Before wastewater can be treated, it must be pumped to the treatment plant. It must then be aerated and filtered. All this requires energy. Exactly how much energy it requires may vary widely, depending on where the treatment plant is located. On average, wastewater treatment in California uses 500–1,500 kilowatt-hours per acre-foot, although this figure can be higher when the

wastewater is pumped uphill for treatment.

WHERE WILL THE WATER COME FROM?

To keep up with growing demand, water agencies across the country are having to implement water efficiency programs and look for new sources of water—generally some mix of surface water, groundwater, recycled water, and desalinated water. For any given location, the energy required by each of these alternatives can vary tremendously. The San Diego County Water Authority, for example, lists the energy requirements for the four alternatives as ranging from less than 100 kilowatt-hours to more than 4,000 kilowatt-hours per acre-foot (see Figure 1). These figures represent the energy required to extract and convey water; they do not include the energy costs of initial treatment, local distribution, end use, or wastewater treatment. What follows is a general look at how the four alternative sources compare when their embedded energy costs are taken into consideration.

Surface water. The energy intensity of surface water depends greatly on the source of that water. In areas where water is gravity fed, no energy may be necessary for conveyance. However, in many parts of the West, water is pumped over long distances. Furthermore, even gravity-fed water is frequently pumped into and out of reservoirs. The amount of energy used to deliver water to households in Southern California, for example, is equivalent to about one-third of total household electric use in the region.

Groundwater. The energy required to extract and deliver groundwater depends on the depth to groundwater, and on the efficiency of the pumps. Along California's central coast and in parts of the San Joaquin basin, groundwater pumping requires on average 292 kilowatt-hours per acre-

foot. In Los Angeles, the figure is 580 kilowatt-hours per acre-foot, and the Yuima Municipal Water District, in San Diego County, reports 661 kilowatt-hours per acre-foot. The Westlands Water District requires 740 kilowatt-hours per acre-foot.

In some regions of the country, groundwater is used faster than it can be replenished. As groundwater is depleted, and depth to groundwater increases, more and more energy is required to pump that water. Groundwater replenishment programs, and concurrent use of surface and groundwater supplies, must be managed with energy costs in mind.

Recycled water. The California Water Code defines recycled water as “water which, as a result of treatment of waste, is suitable for a direct beneficial use or controlled use that would not otherwise occur.” Recycled water is most commonly used for recharging or replenishing groundwater supplies or for landscape or irrigation purposes. While there have been some proposals to reuse recycled water as potable water, no such project has yet been implemented in the United States.

The energy costs of recycling water fall under two heads. The first are the incremental treatment costs required to treat wastewater to the standard necessary for its intended use. The second are the costs of any energy required to convey the water to its place of use. Costs vary depending on the type of project being developed, the degree of treatment required, and the proximity of the water treatment plant to the place where the recycled water will be used. For regions with high energy costs for surface or groundwater supplies, water recycling may be an energy-efficient alternative.

Orange County, California, is constructing a water-recycling system that will use only half the amount of energy required to import the same amount of water from Northern California. According to the Orange County Water District, this water-

recycling system will take highly treated sewer water from the Orange County Sanitation District—water that is currently discharged into the ocean—and will purify it to near-distilled water quality, using high-tech membrane purification systems backed by ultraviolet light with hydrogen peroxide disinfection. After the purification process, the water will be used to expand an existing underground water barrier, which helps prevent seawater from intruding into the groundwater basin. The water will then be pumped into Orange County’s groundwater basin for further natural treatment before it is extracted for later use.

When the project is completed, in 2007, it will produce 70 million gallons of water per day—enough to supply the needs of 144,000 families a year. The water can be purified for less than the cost of imported water, using one-half the energy required to import water from Northern California. This saves an estimated 140 million kilowatt-hours annually—enough energy to serve approximately 21,300 homes each year.

An analysis of water supply alternatives done for the San Diego County Water Authority found that after local surface supplies and conservation, water recycling was the next most energy-efficient source of water supply. Water recycling required 400 kilowatt-hours per acre-foot, compared with 3,240 kilowatt-hours per acre-foot for imported surface supplies from the state water project. A recent analysis by the California Energy Commission estimated that the energy required for water recycling in California ranges from 325 to 1,000 kilowatt-hours per acre-foot.

Recycled water already plays an important role in the water supply of many communities. And the amount of water that is recycled could be vastly expanded. For example, by 2020 more than 3 million acre-feet of wastewater will be generated annually by California’s urban coastal areas. Much of this water could be recycled. Many real and perceived economic

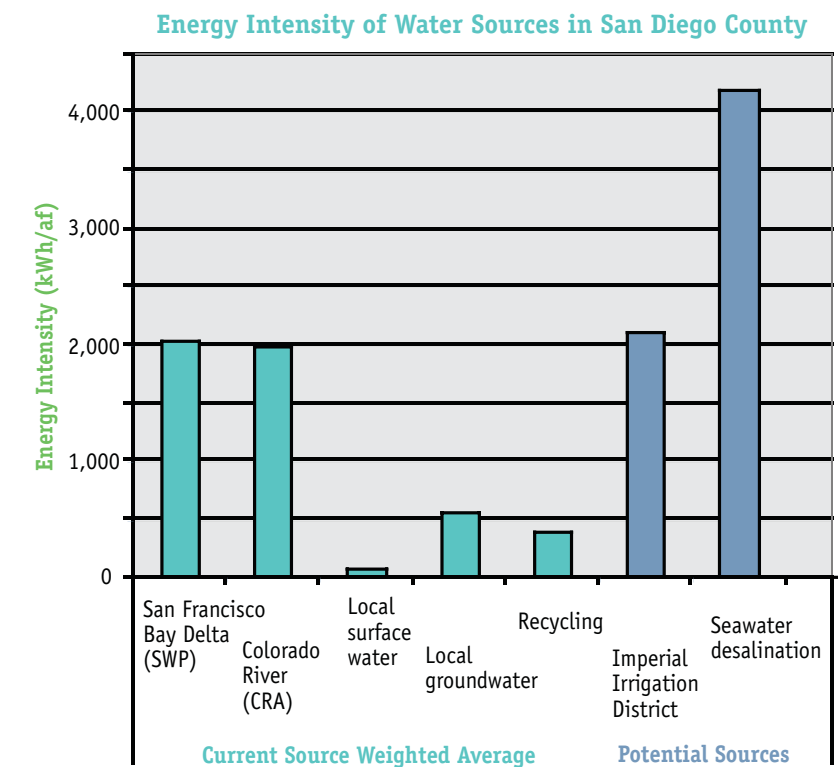


Figure 1. An analysis of water supply alternatives done for the San Diego County Water Authority found that after local surface supplies and conservation, water recycling was the next most energy-efficient source of water supply. Source: Ronnie Cohen, Barry Nelson, and Gary Wolff. *Energy Down the Drain* (New York: Natural Resources Defense Council, 2004) p. 34.

and political barriers still prevent the dramatic expansion of wastewater recycling programs. But if the energy costs of the alternatives are given adequate consideration, we may see more water- recycling projects in the future.

Desalinated water. Ninety-seven percent of the earth’s water is too salty to drink or to irrigate crops. However, salt water can be converted to fresh water through a process called desalination. Desalination has been limited in the United States because it is very expensive; less than 1% of California’s current water supply comes from desalination. However, it is a topic of growing interest as technological advances are reducing desalination costs.

The economics of desalination are directly tied to the cost of energy; energy currently accounts for approximately 40% of total costs. There are various methods of desalination, each of which uses a different amount

of energy. Energy use can vary within a given method as well, depending on the quality of the source water and the details of the desalination design.

The most common choice for new desalination plants today is reverse osmosis, in which salty water is filtered under high pressure through a semipermeable membrane. This method accounts for 90% of California’s current desalination capacity.

Current energy requirements and total costs for desalination vary widely, depending on the specifics of the project. Here are some examples.

- Energy requirements for a seawater desalination plant under investigation by the Municipal Water District of Orange County, California, are estimated at 5,500 kilowatt-hours per acre-foot.
- The Carlsbad seawater desalination project in San Diego County is

estimated to use 5,400 kilowatt-hours per acre-foot.

- Staff of Ionics Corporation reports that the new Trinidad seawater desalting plant will use about 4,800 kilowatt-hours per acre-foot. This is the largest reverse-osmosis facility in the Western Hemisphere, producing 28.8 million gallons per day of potable and industrial grade water.
- A proposed seawater desalination plant serving the Inland Empire Utilities Agency in Southern California would use about 4,400 kilowatt-hours per acre-foot.

A recent paper issued by the Affordable Desalination Collaboration claims that “[by] using commercially available technologies applied in a manner where design emphasis is placed on energy efficiency and responsibly reducing overall total water costs...seawater desalination can now be considered cost-competitive with other new water supply options for California.” This study indicates that power consumption can be less than 2,000 kilowatt-hours per acre-foot for desalination, which would make it competitive, on an energy basis, with other supply sources for the region.

One of the models for lowering energy costs for desalination is colocating facilities with coastal power plants. Many of these power plants have once-through cooling systems, however, which discharge warmer water than they suck in. The warmed water harms aquatic ecosystems that can not adapt to a different temperature. Desalination should not be used to justify extending the life of such a system. In order for desalination projects—including colocated facilities—to gain acceptance, they will need to demonstrate that they reduce the net environmental impact of generating new water supplies, by using the desalinated water to replace freshwater diversions from sensitive ecosystems, for example.

Another way to reduce the energy cost of desalination is to treat brackish groundwater, instead of salty water. This method can produce water at a lower total cost, and a lower energy cost, than ocean desalination. Brackish groundwater treatment may use a great deal of energy or very little, depending on the quality of the source water. Desalting brackish groundwater at the Chino Desalter Facility in southern California requires 1,700 kilowatt-hours per acre-foot, while the Reynolds water treatment plant in San Diego County uses only 405 kilowatt-hours per acre-foot for the same purpose. The same technology can be used to clean up groundwater polluted by other contaminants.

USING WATER EFFICIENTLY

In the past water agencies have treated the efficient use of water as a customer service program, rather than as an alternative source of water. In water-scarce regions, this has started to change; many water agencies are now practicing integrated resources planning, in which efficiency, or demand-side, alternatives are compared to traditional supply-side alternatives as potential ways of meeting a community’s future water needs.

Indeed, the city of Los Angeles, thanks to its water use efficiency programs, is using the same amount of water today as it did 25 years ago, while its population has grown by one million. The Los Angeles Department of Water and Power (LADWP) was one of the first agencies to offer rebates to replace old toilets with ultralow-flush toilets, and in 1992 LADWP began to work with community-based organizations to help distribute fixtures to low-income customers. Since 1999 the city has had a retrofit-upon-resale ordinance that requires that, prior to the close of escrow, residential property owners

- replace all non-water-conserving showerheads with low-flow showerheads; and

- replace all non-water-conserving toilets with ultralow-flush toilets.

The ordinance also requires that nonresidential property owners replace all non-water-conserving showerheads with low-flow showerheads and install flush reduction devices in all non-water-conserving toilets.

LADWP also offers rebates for high-efficiency clothes washers, and a variety of measures for businesses and multifamily building owners, including weather-based irrigation controllers, cooling-tower conductivity controllers, x-ray processor recirculating systems, and other measures. LADWP offers commercial, industrial, institutional, and multifamily building owners in Los Angeles up to \$100,000 for the installation of preapproved equipment and products.

A recent analysis of the energy cost of water conducted by the San Diego County Water Authority found that end use represents the single largest component of water-related energy cost. If this is true for regions with energy-intensive sources of water, like San Diego, where the energy cost of conveyance is expected to dwarf the cost of treatment, distribution, and so forth, then it is likely to be even truer for other regions. This suggests that there is a potential for enormous energy savings to be realized by using water more efficiently.

Residential water use accounts for 50%–85% of urban water use. Nationally, single-family residential demand averages 101 gallons per capita daily, and multifamily residential demand averages 45–70 gallons per capita daily. Using water more efficiently may be the single best way to reduce water-related energy costs, because using water more efficiently has many benefits. It saves the on-site energy; it saves the upstream energy required to extract, convey, treat, and distribute the water; and it also saves the downstream energy needed to treat and dispose of wastewater.

There are many ways to use water more efficiently, and factoring energy benefits into the analysis of these

ways boosts their cost-effectiveness. Efficiency measures that reduce end use energy include installing efficient showerheads, dishwashers, and clothes washers. (See “Saving Water Indoors,” p. 26.) Federal and state programs to promote market transformation, along with state and local rebate programs, can accelerate the adoption of these measures.

COLD WATER SAVINGS

The largest two uses of water in the residential sector are toilets and landscaping. These do not require significant end use energy, since they use cold water. However, even cold water conservation can save energy. According to recent estimates by the California Energy Commission, the energy costs for water supplied for outdoor uses average 3,500 kWh per acre foot in Northern California and over 11,000 kWh per acre foot in Southern California. When wastewater treatment is added into the equation, those numbers jump to 5,400 kWh per acre foot in Northern California and over 13,000 kWh per acre foot in Southern California.

Toilet flushing represents the largest single use of water inside the home. While older toilets use as much as 6 gallons per flush, federal and state water efficiency laws now standardize flush volumes at a maximum of 1.6 gallons per flush for all new toilets. Many water utilities in the United States have conserved a significant amount of water by installing ultralow-flush toilets. However, there are significant additional savings to be realized from ultralow-flush toilet retrofits, and from the newer generation of high-efficiency toilets (see “The Real High-Efficiency Toilets Have Arrived,” p. 10).

Outdoor landscaping represents about 30% of residential water use nationwide. In some hot, dry areas, landscape demand can shoot up to 75% of residential use. Yet most irrigation systems don’t use water efficiently. Leaking valves and broken sprinkler heads may go unnoticed, and overwatering can waste much water. In

addition to wasting water, overwatering may cause runoff into storm gutters, carrying residues from lawn and garden chemicals and fertilizers into bays, rivers, and streams.

In the past, water agencies have focused their conservation programs primarily on indoor water use, but a new generation of programs have



To supply water to Southern California, the State Water Project pumps water over the 2,000-ft. Tehachapi Mountains—the highest lift of any water system in the world.

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begun to target outdoor water use, using such tools as dedicated landscape meters—“smart” controllers that adjust landscape irrigation based on weather conditions. (See “Smart Irrigation,” p. 24.) Studies have shown that it is possible to reduce outdoor water use by 25%–50% using currently available technologies. Furthermore, landscape water use peaks during the hot summer months when energy use is also peaking. Reducing landscape water use can thus help to alleviate pressure on the power grid.

One effective way to target the overuse of water for landscaping might be to charge more for it. Studies have shown that people are less willing to pay a high price for water used outdoors than for water used indoors.

Charging higher rates for water that is used for landscaping might encourage people to use that water more efficiently.

AGENCIES TAKING STRIDES

Some water agencies have become aware of the energy–water connection, and have begun to implement policies to reduce their use of energy. Some energy utilities are likewise becoming aware of this connection.

Inland Empire Utilities Agency.

The Inland Empire Utilities Agency (IEUA) serves a rapidly growing population in San Bernardino County, in Southern California. This agency supplies its customers with more than 200,000 acre-feet of water a year. The water that it imports from the east branch of the state water project requires 3,200 kilowatt-hours per acre-foot for conveyance alone; once treatment and local distribution are added in, it requires almost 3,400 kilowatt-hours per acre-foot—all this before end use and wastewater treatment. After the California energy crisis of 2000–01, IEUA adopted a policy of energy self-sufficiency and evaluated its options for providing water in light of the energy that each option requires. This led the agency to adopt an integrated water management strategy that reduces dependence on high-energy water supplies by establishing aggressive efficient-use programs, and by developing water-recycling, groundwater, and storm water recapture programs.

Before IEUA adopted this strategy, it expected imported water supplies to increase by 90,000 acre-feet—roughly 40%—a year. Under the new strategy, IEUA now expects to decrease imported water supplies from current levels, while experiencing a population growth of over 300,000, or more than 35%. IEUA’s water-recycling program is expected to produce approximately 100,000 acre-feet a year, replacing the same amount of imported water. This translates to savings of 34 megawatts a year. The development

and use of recycled water within IEUA's service area will also reduce greenhouse gas emissions by 100,000 tons of carbon dioxide equivalents per year. "With only 10% of the total available wastewater being recycled in Southern California, there is huge potential for additional energy savings and greenhouse gas reductions from aggressive development of recycled water supplies," says Wyatt Troxell, IEUA board president.

Santa Clara Valley Water District. Located in San Jose, California, the Santa Clara Valley Water District (SCVWD), provides water to 1.8 million residents. To help meet the growing demand for water, SCVWD has developed a suite of efficient-use programs that saved 315,800 acre-feet of water between 1993 and 2005. During this 12-year period, SCVWD estimates that its water conservation programs saved approximately 1.4 billion kilowatt-hours of energy. This represents a financial savings of about \$190 million and is equivalent to a year's worth of electricity for over 200,000 households. SCVWD estimates that by saving this energy, it eliminated the discharge of approximately 335 million kilograms of carbon dioxide. This is the equivalent of removing 72,000 cars from the road for one year. "We're not the only water agency that promotes water use efficiency, but we may be one of the first to quantify how our water conservation efforts save energy and reduce carbon dioxide emissions," says Hossein Ashktorab, who manages water use efficiency programs at the district.

Energy utilities. Just as some water agencies are recognizing the energy-saving benefits of conserving water, so some energy utilities in California are seeing water conservation as a new way to save energy. This method got increased attention after a report from the California Energy Commission suggested that conserving water might be a more cost-effective approach to saving energy than traditional energy-saving programs. Preliminary estimates

showed that California could save 95% of the energy saved by implementing the usual energy efficiency programs at only 58% of the cost by conserving water. Although these are not the final figures, they suggest that energy utilities and water agencies might well collaborate to help conserve these two precious and connected resources.

CLIMATE CONCERNS

Any choice of new water supplies, or reexamination of existing supplies, must factor in the future impacts of climate change on those alternatives. Scientists predict that global warming will alter precipitation patterns; decrease snow pack; increase winter stream flows; and result in hotter, dryer summers. Any one of these changes will affect the water supply. Taken together, they mean that we are facing a future that will be vastly different from our past.

A remarkable number of studies and reports reflect a growing consensus within the scientific community regarding climate change, and an improved understanding of its implications for water management. A U.S. National Assessment water sector report noted that "more than twenty years of research and more than a thousand peer-reviewed scientific papers have firmly established that a greenhouse warming will alter the supply and demand for water, the quality of water, and the health and functioning of aquatic ecosystems."

Climate change creates a lot of uncertainty about future water supplies. Old-style solutions, such as constructing new dams, are likely to be less cost-effective under most climate change scenarios, partly because the yield of the dam will be reduced if precipitation decreases, and partly because more of the reservoir will have to be kept empty for flood control purposes. The approaches that were identified above as saving the most energy—efficient-use programs and recycling—are also likely to be the best performers in the uncertain conditions created by climate change. Water

conservation and water recycling offer a two-for-one approach; they can help water agencies to meet the demand for water under a variety of climate change scenarios, while simultaneously helping them to save energy, thereby reducing the emissions that contribute to climate change.

As the water-energy nexus starts to get more attention, more people will begin to recognize the role that improved conservation, recycling, and other water management alternatives can play in saving energy. When it comes to saving energy, turning off the tap is like turning off the lights.

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For more information:

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For more information on the recycled water project in Orange County, go to www.gwrsystem.com/news/releases/050209.html.

For more information on the Santa Clara Valley Water District initiatives, see www.valleywater.org.

Navigant Consulting, Inc. 2006. *Refining Estimates of Water-Related Energy Use in California*. California Energy Commission, PIER Industrial/Agricultural/Water End Use Energy Efficiency Program. CEC-500-2006-118. To download this report, go to www.energy.ca.gov/pier/final_project_reports/CEC-500-2006-118.html.